

Pairing of universes of masses $+m$ and $-m$: exact proofs for dirac16complex and dirac16complex00

The pairing theorems T1, T2 and T3 and the quantum-level reading in the author's
primordial gravitational field, with every hypothesis, the verification records and an exact
statement of what they do not establish

dirac16complex project (with Claude Opus 5.5)

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Abstract

The author asked on 2026-10-01: “PROVE that Universes of masses $\{+mass, -mass\}$ are created in pairs for each case of the `dirac16complex` and the `dirac16complex00` fields.” This document gives the exact answer computed by the code in `Revision/` for the author’s primordial metric. Proved, for both fields and with every hypothesis stated: T1 (chirality pairing): the chirality matrix Γ maps every configuration Ψ of the theory with mass and coupling (m, λ) to the configuration $\Gamma\Psi$ of the theory with $(-m, -\lambda)$, with $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$, solutions to solutions and the energy-momentum tensor and the current reversed, so that the pair has zero total energy-momentum, current and charge as classical bilinears, in every gravitational field. T2 (mirror pairing): Γ combined with a $\text{Pin}(4,4)$ reflection of character -1 maps (m, λ) to $(-m, \lambda)$ with $\mathcal{L} \rightarrow +\mathcal{L}$; in the author’s field it is the mirror across the Z2 brane $z = \pi/2$ (an ASSUMED construction), and the partner has EQUAL, not opposite, energy-momentum. T3 (Kohn-Sham level, `dirac16complex`): the block map of Γ , with the brane parities exchanged and the tip angle $\theta \rightarrow \pi - \theta$, maps every self-consistent instantaneous Kohn-Sham state with (m, λ, θ) onto one with $(-m, +\lambda, \pi - \theta)$, with equal levels, occupations, Kohn-Sham energies and energy-momentum profiles (ASSUMED Z2 brane, mean-field level); a completion of 2026-10-08 adds the 16-component statement S6 (every component of the energy-momentum tensor and of the current is unchanged, S and Q change sign), carries the filling convention onto the $-m$ member and demonstrates T3 numerically with two independent Kohn-Sham solvers (demonstrations, not part of the proof). The quantum-level reading for `dirac16complex`: the chirality image carries the Krein metric $-B$ and is the same quantum system re-labelled; two independently quantised universes of masses $+m$ and $-m$ have identical one-particle ($\lambda = 0$) spectra (flat space, or frozen coefficients at a point) and their generators do not cancel. A corollary: a T1 pair taken as the complete classical source of the author’s metric is a zero source, and the Einstein equations then have no solution for $H > 0$. Not established by any of these equations: a creation process, a rate, a probability or an amplitude for creating universes, in pairs or otherwise. T1, T2, T3 and the quantum reading Q are proved, each under its stated hypotheses; the creation of pairs is not.

1 The request and the exact answer

The author’s request, quoted from `Revision/README.md`: “PROVE that Universes of masses $\{+mass, -mass\}$ are created in pairs for each case of the `dirac16complex` and the `dirac16complex00` fields.”

What a universe of mass m means in this record: a configuration of one of the two fields with mass parameter m and coupling λ in the author’s primordial gravitational field, i.e. a solution of the Euler-Lagrange equations of section 2.4. Two such universes are paired when an explicit invertible map takes every solution of the theory with mass m to a solution of the theory with mass $-m$ and takes its Lagrangian, energy-momentum tensor and current to stated multiples of those of the partner.

The exact answer, for both fields:

1. Proved (sections 4 to 8): the pairing theorems T1 and T2, the quantum-level reading Q for `dirac16complex`, the corollary C1 for the field equations of a_4 and the Kohn-Sham-level theorem T3 for `dirac16complex`, each with its exact hypotheses (T2 and T3 use the ASSUMED Z2 construction at the brane).
2. Not proved, and not derivable from these equations: that such universes are CREATED. The equations are field equations on a fixed gravitational background and their canonical

quantisation; no equation of this record produces a pair of universes from anything, and no creation process, rate or amplitude follows from them (section 11).

3. The Kohn-Sham level T3 of `Revision/SPEC.md` section 9 is proved in section 8 for the instantaneous (adiabatic) mean-field Kohn-Sham states of `Revision/kohn_sham/ks-theory.json`; the time-dependent Kohn-Sham problem is open. Its completion (section 8.5) closes three gaps found by an adversarial verification and adds two numerical demonstrations.

Every formula and number below is taken from the Revision reports listed in section 9; nothing is taken from the earlier stages of the repository. Interpretations are labelled as such.

Related Revision documents. The Kohn-Sham model of `dirac16complex` in the deflating field, whose states T3 maps, is described in `Revision/docs/KOHN_SHAM_DEFLATING_FIELD` (block reduction, instantaneous ground and first excited states along the prescribed history $a_4 = AHx_4$, Mermin thermodynamics, the full cross-check of the Rust solver against the independent reference, 31 of 31 checks). The Lovelock tensors behind the field equations for a_4 used by corollary C1 are described in `Revision/docs/LOVELOCK_GKD`. The field equations for a_4 with the Kohn-Sham source (`Revision/field_equations_a4/ks_source`) show that no recorded Kohn-Sham state is an admissible source of the author's metric (section 8.4), so T3 is a statement about test-field states on a prescribed background. The dark-sector hypotheses for both fields are investigated in `Revision/docs/DARK_SECTOR_HYPOTHESES`; that record establishes neither Hypothesis nor Hypothesis00 (it states what each field can and cannot produce under its stated observer assumption), and nothing in it is used by, or changes, the theorems of this document.

2 Setting and conventions

2.1 The author's metric and coordinates

The coordinates are named as the author names them: x_1, x_2, x_3 are ordinary 3-space (inflating), x_4 is the time, x_5, x_6, x_7 are the three extra times, which DEFLATE exponentially as a_4 increases, and x_8 is the hidden space direction, with $z = 6Hx_8 \in (0, \pi/2)$ and $H > 0$ the author's constant. The metric of `Revision/SPEC.md` section 1 is

$$ds^2 = e^{2a_4(x_4)} \sin^{1/3} z (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \\ - e^{-2a_4(x_4)} \sin^{1/3} z (dx_5^2 + dx_6^2 + dx_7^2) + \cot^2 z dx_8^2,$$

of signature (4,4) (space-like x_1, x_2, x_3, x_8 ; time-like x_4, x_5, x_6, x_7), with $\sqrt{|g|} = \cos z$. The diagonal vielbein is $e^a{}_\mu = f_a \delta^a_\mu$ with $f_1 = f_2 = f_3 = e^{a_4} \sin^{1/6} z$, $f_4 = 1$, $f_5 = f_6 = f_7 = e^{-a_4} \sin^{1/6} z$ and $f_8 = \cot z$; the frame metric is $\eta = \text{diag}(+1, +1, +1, -1, -1, -1, -1, +1)$ in the order $x_1, \dots, x_8$. Frame indices $a = 1, \dots, 8$ are aligned with the coordinates, so γ^4 is the time gamma $\gamma^{(x_4)}$ and γ^8 the hidden-direction gamma $\gamma^{(x_8)}$.

The canonical spin connection (vielbein postulate) has the nonzero components $\omega_{x_i, i4} = a'_4 e^{a_4} \sin^{1/6} z$ and $\omega_{x_i, i8} = H e^{a_4} \sin^{1/6} z$ for the inflating directions $i = 1, 2, 3$, and $\omega_{x_t, 4t} = -a'_4 e^{-a_4} \sin^{1/6} z$ and $\omega_{x_t, t8} = -H e^{-a_4} \sin^{1/6} z$ for the deflating extra times $t = 5, 6, 7$, with $\Omega_{x_4} = \Omega_{x_8} = 0$. Contracted with the gammas, $\gamma^\mu \Omega_\mu = 3H\gamma^8$: each inflating direction contributes $\frac{a'_4}{2}\gamma^4 + \frac{H}{2}\gamma^8$ and each deflating extra time $-\frac{a'_4}{2}\gamma^4 + \frac{H}{2}\gamma^8$, so the time-direction terms cancel and the hidden-direction terms add.

2.2 Clifford data

The gammas are the author's real 16×16 matrices T16, re-constructed in Revision code from the author's tau formulas, with $\gamma^8 = T16[0]$, $\gamma^{1,2,3} = T16[1..3]$, $\gamma^4 = T16[4]$ and $\gamma^{5,6,7} = T16[5..7]$. They are real signed permutation matrices with $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$ and $(\gamma^a)^T = \eta_{aa}\gamma^a$. Further:

- $C = \gamma^8\gamma^1\gamma^2\gamma^3$ is real and symmetric, $C^2 = 1$, $C\gamma^a$ is real and antisymmetric and $C\gamma^a C^{-1} = -(\gamma^a)^T$; the Dirac adjoint is $\bar{\Psi} = \Psi^\dagger C$.
- $\Gamma = \gamma^8\gamma^1\gamma^2\gamma^3\gamma^4\gamma^5\gamma^6\gamma^7 = \text{diag}(-I_8, I_8)$.
- $B = -iC\gamma^4$ is Hermitian with $B^2 = 1$ and signature (8,8).
- $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ satisfy the commutation relations of $\text{so}(4,4)$.
- The 16-dimensional representation is irreducible under $\text{Pin}(4,4)$ (commutant of dimension 1) and splits under $\text{Spin}(4,4)$ into two inequivalent irreducible 8-dimensional representations, the chiral halves $\Gamma = -1$ (components 1 to 8, written ψ_-) and $\Gamma = +1$ (components 9 to 16, written ψ_+); every γ^a exchanges the two halves.

2.3 The two fields and the Lagrangian

dirac16complex is Ψ : 16 complex ANTICOMMUTING (Grassmann) components, a $\text{Pin}(4,4)$ spinor, quantised canonically (section 6). dirac16complex00 is Φ : 16 complex COMMUTING components that transform as a $\text{Pin}(4,4)$ spinor, a classical field. Both are coupled to gravity through the vielbein and the canonical spin connection. The theorems are stated with the letter Ψ for both fields; “both statistics” means that the statement and its proof hold for Grassmann components and for commuting components. With $S = \bar{\Psi}\Psi$, $D_\mu\Psi = \partial_\mu\Psi + \Omega_\mu\Psi$, $D_\mu\bar{\Psi} = \partial_\mu\bar{\Psi} - \bar{\Psi}\Omega_\mu$, $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$ and $\gamma^\mu = e^\mu{}_a\gamma^a$, the Lagrangian density of both fields is

$$\mathcal{L}_{m,U}[\Psi] = \sqrt{|g|} [K[\Psi] - mS - U(S)], \quad K[\Psi] = \frac{1}{2}(\bar{\Psi}\gamma^\mu D_\mu\Psi - (D_\mu\bar{\Psi})\gamma^\mu\Psi),$$

written $\mathcal{L}_{m,\lambda}$ for $U(S) = \frac{\lambda}{2}S^2$. It is real for both statistics.

2.4 Field equation, energy-momentum tensor, current

The Euler-Lagrange equations of $\mathcal{L}_{m,U}$ are, for both statistics,

$$E_{m,U}[\Psi] \equiv \gamma^\mu D_\mu\Psi - (m + U'(S))\Psi = 0, \quad (D_\mu\bar{\Psi})\gamma^\mu = -(m + U'(S))\bar{\Psi}.$$

The symmetric (Belinfante) energy-momentum tensor of the theory record is

$$T^\nu{}_\mu = \delta^\nu{}_\mu \frac{\mathcal{L}}{\sqrt{|g|}} - \frac{1}{4} \left(\bar{\Psi}\gamma^\nu D_\mu\Psi - (D_\mu\bar{\Psi})\gamma^\nu\Psi + \bar{\Psi}\gamma_\mu D^\nu\Psi - (D^\nu\bar{\Psi})\gamma_\mu\Psi \right),$$

with the energy density $\rho = -T^{x_4}_{x_4}$ and the pressures $p_\mu = T^\mu_\mu$ (no sum) of `Revision/SPEC.md` section 4; for homogeneous on-shell states $\rho = mS + U$ and $p = SU' - U$. The conserved current is $J^\mu = -i\bar{\Psi}\gamma^\mu\Psi$, with the charge density $J^{x_4} = \Psi^\dagger B\Psi$ and the charge $Q = \int \cos z \Psi^\dagger B\Psi d^7x$.

Convention note. The pairing records use the tensor

$$T_{\mu\nu} = \frac{1}{4}(\bar{\Psi}\gamma_\mu D_\nu\Psi + \bar{\Psi}\gamma_\nu D_\mu\Psi - (D_\mu\bar{\Psi})\gamma_\nu\Psi - (D_\nu\bar{\Psi})\gamma_\mu\Psi) - g_{\mu\nu}\mathcal{L}/\sqrt{|g|},$$

which is minus the tensor above, and the currents $\bar{\Psi}\gamma^\mu\Psi$ (Wolfram) and $i\bar{\Psi}\gamma^\mu\Psi$ (sympy). Every statement of T1, T2 and Q about T and J is linear and homogeneous in them ($T' = \pm T$, $J' = \pm J$), so it holds unchanged for every overall sign and constant factor.

2.5 The field equation in the author's metric

Written out in the author's metric, the field equation of both fields is

$$e^{-a_4} \sin^{-1/6} z (\gamma^1 \partial_1 + \gamma^2 \partial_2 + \gamma^3 \partial_3) \Psi + \gamma^4 \partial_4 \Psi + e^{a_4} \sin^{-1/6} z (\gamma^5 \partial_5 + \gamma^6 \partial_6 + \gamma^7 \partial_7) \Psi \\ + \tan z \gamma^8 \partial_8 \Psi + 3H\gamma^8 \Psi = (m + U'(S)) \Psi,$$

where $3H\gamma^8 = \gamma^\mu \Omega_\mu$ is the spin-connection term; it is nonzero for every $H > 0$ and every $\Psi \neq 0$ (non-triviality [1] and [2] of the theory record). In the chiral block form $\Psi = (\psi_-, \psi_+)$ every gamma is block off-diagonal, with upper-right block $\bar{\tau}^a$ and lower-left block τ^a , and $\bar{\tau}^8 = \tau^8 = I_8$; the 16 equations become the two 8-component equations

$$\sum_a f_a^{-1} \bar{\tau}^a \partial_a \psi_+ + 3H\psi_+ = V\psi_-, \quad \sum_a f_a^{-1} \tau^a \partial_a \psi_- + 3H\psi_- = V\psi_+, \quad V = m + U'(S).$$

2.6 Records for section 2

item	Wolfram records	sympy records
metric, vielbein, $\sqrt{ g } = \cos z$	<code>metric_is_the_authors,</code> <code>sqrt_det_g_is_cos_z,</code> <code>primordial_vielbein</code>	<code>metric_from_viel</code> <code>bein_equals_SPEC,</code> <code>sqrt_det_g_equals_cos_z</code>
spin connection and $\gamma^\mu \Omega_\mu = 3H\gamma^8$	<code>omega_components,</code> <code>Omega_components,</code> <code>gammaOmega_equal</code> <code>s_3H_gamma_x8, gammaOme</code> <code>ga_x4_terms_cancel</code>	<code>geometry.spin_co</code> <code>nnection_components,</code> <code>gamma_mu_Omega_m</code> <code>u_equals_3H_gamma_x8,</code> <code>time_terms_cance</code> <code>l_hidden_term_survives</code>
gammas	<code>coordinate_map, Clifford</code> <code>_relation, reality,</code> <code>signed_permutati</code> <code>on_matrices, symmetry</code> <code>_pattern</code>	<code>clifford_relation,</code> <code>reality_signed_p</code> <code>ermutations, symmetry</code> <code>_pattern</code>

item	Wolfram records	sympy records
C, Γ, B, S^{ab}	C_real_symmetric, C_squared_identity, C_gamma_real_antisymmetric, C_gamma_inverse, Gamma_definition, Gamma_diag, B_Hermitian, B_square d_identity, B_signature_8_8, S_Lorentz_algebra	C_real_symmetric_involution, C_gamma_antisymmetric, C_conjugation, chirality_diag, B_hermitian_involution_signature, S_lorentz_algebra
Pin(4,4) and Spin(4,4)	Pin44_irreducible_commutant_dim_1, Spin44_commutant_dim_2_chiral_projectors, chiral_halves_irreducible, chiral_halves_in_equivalent_intertwiners_0, Spin_Pin_reflection_swaps_halves	pin_commutant_dimension_1, spin_commutant_dimension_2, spin_halves_irreducible, spin_halves_inequivalent, reflections_exchange_halves
reality of $\mathcal{L}$	L_real_G, L_real_C	grassmann_lagrangian_real, commutator_lagrangian_real
Euler-Lagrange equations	EL_Psi_G, EL_Psi_bar_G, EL_Psi_C, EL_Psi_bar_C, T1_Euler_Lagrange_derived_primordial_grassmann, T1_Euler_Lagrange_derived_primordial_commuting	grassmann_euler_lagrange_psi_variation, grassmann_euler_lagrange_psi_bar_variation, commutator_euler_lagrange_psi_variation, commutator_euler_lagrange_psi_bar_variation
energy-momentum tensor	T_symmetric_part_Belinfante_G, T_symmetric_part_Belinfante_C, kinetic_sum_on_shell_G, kinetic_sum_on_shell_C	grassmann_emt_symmetric, commuting_emt_symmetric, grassmann_homogeneous_on_shell_rho_p, commuting_homogeneous_on_shell_rho_p
current and charge	current_conservation_identity_G, current_conservation_identity_C, charge_density_is_Krein_form_G, charge_density_is_Krein_form_C	grassmann_current_conservation, commutator_current_conservation

item	Wolfram records	sympy records
field equation in the metric, block form, non-triviality	Dirac_operator_e xplicit_G, Dirac_op erator_explicit_C, block_form, nontrivi ality_1_dirac16complex, nontriviality_2_ dirac16complex00	nontriviality_0m ega_zero_iff_flat

3 Algebraic lemmas

All four lemmas are exact matrix identities of the Revision gammas, verified by WolframScript and independently by sympy (records after each proof).

Lemma 1 (the chirality matrix). Γ is real and symmetric, $\Gamma^2 = 1$, $\Gamma\gamma^a = -\gamma^a\Gamma$ for every a , $\Gamma C = C\Gamma$ and $\Gamma S^{ab} = S^{ab}\Gamma$. Hence $\Gamma\Omega_\mu = \Omega_\mu\Gamma$ for EVERY connection $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$, and $D_\mu(\Gamma\Psi) = \Gamma D_\mu\Psi$, $D_\mu(\bar\Psi\Gamma) = (D_\mu\bar\Psi)\Gamma$.

Proof. Γ is the product of all eight different γ^a . Moving γ^b through it passes the seven factors γ^a , $a \neq b$, which anticommute with γ^b , and the factor γ^b itself, which commutes with it: the sign is $(-1)^7 = -1$. C is a product of four and S^{ab} ($a \neq b$) a multiple of a product of two different gammas, so both commute with Γ . $\Gamma = \text{diag}(-I_8, I_8)$ is computed exactly; it is real, symmetric and squares to 1. Γ is constant and commutes with Ω_μ , which gives the statements about D_μ . QED.

Wolfram records	sympy records
Gamma_properties, Gamma_diag, Gamma_anticommutates_with_gammas, Gamma_commutates_with_C, S_commut es_with_Gamma	gammas.Gamma, gammas.Gamma_ant icommutates, chirality_anticommutates, chirality_C_relation, S_preser ves_C_and_commutates_with_Gamma

Lemma 2 (bilinears under the chirality map). For $\Psi' = \Gamma\Psi$: $\bar\Psi' = \bar\Psi\Gamma$; $\bar\Psi'M\Psi' = (-1)^k\bar\Psi M\Psi$ for every product M of k gammas; in particular S is invariant and every bilinear with γ^a , $\gamma^a S^{bc}$ or $S^{bc}\gamma^a$ changes sign. Moreover $\Gamma B\Gamma = -B$.

Proof. By Lemma 1, $\bar\Psi' = \Psi^\dagger\Gamma^\dagger C = \Psi^\dagger\Gamma C = \Psi^\dagger C\Gamma = \bar\Psi\Gamma$. Then $\bar\Psi'M\Psi' = \bar\Psi\Gamma M\Gamma\Psi$, and $\Gamma M\Gamma = (-1)^k M$ by Lemma 1. Finally $\Gamma B\Gamma = -iC\Gamma\gamma^4\Gamma = iC\gamma^4 = -B$. QED.

Wolfram records	sympy records
T1_kernel_scalar, T1_kernel_kinetic, T1_kernel_connection (all 512 triples), T1_kernel_field_equation, Q_Krein_ metric_of_images	T1.general_field.matrix_identities (456 kinetic and connection basis matrices), Q.image_krein_metric

Lemma 3 (reflections and their Pin(4,4) lifts). Let n be a frame direction and R_n the diagonal matrix with -1 in place n and $+1$ elsewhere (the reflection of that direction; $R_n\eta R_n = \eta$).

1. $(\gamma^n)^T C \gamma^n = -\eta_{nn} C$.

2. $\gamma^n \gamma^a (\gamma^n)^{-1} = -(R_n)^a_b \gamma^b$ and $\gamma^n S^{ab} (\gamma^n)^{-1} = (R_n)^a_c (R_n)^b_d S^{cd}$.
3. $P_n := \Gamma \gamma^n$ is plus or minus the product of the seven other gammas, hence an element of $\text{Pin}(4,4)$; $P_n^{-1} \gamma^a P_n = (R_n)^a_b \gamma^b$ (it covers R_n); $P_n^\dagger C P_n = \chi_n C$ with the character $\chi_n = -\eta_{nn}$, i.e. -1 for the space-like x_1, x_2, x_3, x_8 and $+1$ for the time-like x_4, x_5, x_6, x_7 ; and $\Gamma P_n = \gamma^n$.

Proof. (1) From $C \gamma^n C^{-1} = -(\gamma^n)^T$ we have $C \gamma^n = -(\gamma^n)^T C$, so

$$(\gamma^n)^T C \gamma^n = -((\gamma^n)^T)^2 C = -((\gamma^n)^2)^T C = -\eta_{nn} C.$$

(2) γ^n anticommutes with γ^a for $a \neq n$ and commutes with itself; the second identity follows by inserting the first into $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$. (3) $\Gamma \gamma^n$ is, up to the sign of the reordering, the product of the eight gammas with γ^n moved to the end, i.e. $\pm \eta_{nn}$ times the product of the other seven; each γ^a is a unit vector ($(\gamma^a)^2 = \eta_{aa} = \pm 1$), so the product lies in $\text{Pin}(4,4)$. Using (2), $(\gamma^n)^{-1} = \eta_{nn} \gamma^n$, Lemma 1 and (1),

$$\begin{aligned} P_n^{-1} \gamma^a P_n &= (\gamma^n)^{-1} \Gamma \gamma^a \Gamma \gamma^n = -(\gamma^n)^{-1} \gamma^a \gamma^n = (R_n)^a_b \gamma^b, \\ P_n^\dagger C P_n &= (\gamma^n)^T \Gamma C \Gamma \gamma^n = (\gamma^n)^T C \gamma^n = -\eta_{nn} C, \end{aligned}$$

and $\Gamma P_n = \Gamma^2 \gamma^n = \gamma^n$. QED. The signs of P_n relative to the product of the other seven gammas (in the order of Γ) are $+1, -1, +1, +1, -1, +1, -1, -1$ for $n = x_1, \dots, x_8$.

Wolfram records	sympy records
T2_Pn_in_Pin44, T2_Pn_covers_the_reflection, T2_character_of_Pn, T2_Gamma_times_Pn_is_gamma_n, T2_kernels_gamma_n_with_frame_reflection	T2.general_field.matrix_identities, T2.general_field.reflection_table, compare.theory.reflection_table

Lemma 4 (Krein signs of the maps). $MBM^\dagger = \sigma_M B$ with $\sigma_\Gamma = -1$; $\sigma_{\gamma^n} = +1$ for $n = x_1, x_2, x_3, x_4, x_8$ and -1 for $n = x_5, x_6, x_7$; and $\sigma_{P_n} = -\sigma_{\gamma^n}$, i.e. -1 for $P_{x_1}, P_{x_2}, P_{x_3}, P_{x_4}, P_{x_8}$ and $+1$ for $P_{x_5}, P_{x_6}, P_{x_7}$.

Proof. Γ : Lemma 2. For γ^n : it is real, so $(\gamma^n)^\dagger = (\gamma^n)^T = \eta_{nn} \gamma^n$, and transposing $C \gamma^n C^{-1} = -(\gamma^n)^T$ with $C^T = C = C^{-1}$ gives $\gamma^n C = -C (\gamma^n)^T$. Hence

$$\gamma^n B (\gamma^n)^\dagger = -i \gamma^n C \gamma^4 (\gamma^n)^T = i C (\gamma^n)^T \gamma^4 (\gamma^n)^T = i C \gamma^n \gamma^4 \gamma^n.$$

For $n \neq 4$, $\gamma^n \gamma^4 \gamma^n = -\eta_{nn} \gamma^4$ and the result is $\eta_{nn} B$; for $n = 4$, $\gamma^4 \gamma^4 \gamma^4 = -\gamma^4$ and the result is B . Finally $P_n B P_n^\dagger = \Gamma \gamma^n B (\gamma^n)^\dagger \Gamma = -\sigma_{\gamma^n} B$ by Lemma 2. QED.

Wolfram records	sympy records
Q_Krein_metric_of_images	compare.theory.krein_signs, Q.image_krein_metric, Q.T2_image_keeps_B

4 Theorem T1: the chirality pairing (both fields)

4.1 Hypotheses

- (H1) The gravitational field: any vielbein $e^a{}_\mu$ (metric $g = e^T \eta e$ of signature (4,4)) and any spin connection $\omega_{\mu ab} = -\omega_{\mu ba}$, in particular the canonical one of the author's metric. The field is a fixed (test) background: it is NOT varied, and both members of the pair are evaluated in the SAME field.
- (H2) The statistics: Ψ Grassmann-valued (dirac16complex) or commuting (dirac16complex00). The map acts at the same point, $\Psi'(x) = \Gamma\Psi(x)$, with no change of coordinates.
- (H3) The potential: $U(S) = \frac{\lambda}{2}S^2$; more generally any function U (commuting field) or any polynomial U (Grassmann field), mapped to $-U$.
- (H4) $\mathcal{L}$, E , T and J as in section 2, with any overall sign or constant factor of T and J .

4.2 Statement

Theorem T1. Under (H1) to (H4), for both fields:

- (T1a) $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$ identically (off shell); for a general potential $\mathcal{L}_{m,U}[\Gamma\Psi] = -\mathcal{L}_{-m,-U}[\Psi]$. Separately, $S[\Gamma\Psi] = S[\Psi]$ and $K[\Gamma\Psi] = -K[\Psi]$.
- (T1b) $E_{m,\lambda}[\Gamma\Psi] = -E_{-m,-\lambda}[\Psi]$: $\Gamma\Psi$ solves the field equations with (m, λ) if and only if Ψ solves those with $(-m, -\lambda)$; equivalently, Ψ solves the (m, λ) equations if and only if $\Gamma\Psi$ solves the $(-m, -\lambda)$ equations. The Euler-Lagrange expressions are mapped in the same way.
- (T1c) $T_{\mu\nu}^{(-m,-\lambda)}[\Gamma\Psi] = -T_{\mu\nu}^{(m,\lambda)}[\Psi]$ and $J^\mu[\Gamma\Psi] = -J^\mu[\Psi]$ pointwise, off and on shell.
- (T1d) The pair $(\Psi \text{ with } (m, \lambda); \Gamma\Psi \text{ with } (-m, -\lambda))$ has the total energy-momentum density $T + T' = 0$, the total current $J + J' = 0$ and the total charge $Q + Q' = 0$, as classical bilinears (c-numbers for dirac16complex00, elements of the Grassmann algebra for dirac16complex).

4.3 Proof

Step 1 (bilinears). By Lemma 1, $D_\mu(\Gamma\Psi) = \Gamma D_\mu\Psi$ and $D_\mu(\bar{\Gamma}\bar{\Psi}) = (D_\mu\bar{\Psi})\bar{\Gamma}$; by Lemma 2, $S[\Gamma\Psi] = S[\Psi]$ and

$$\bar{\Gamma}\bar{\Psi}\gamma^\mu D_\mu(\Gamma\Psi) = \bar{\Psi}\Gamma\gamma^\mu\Gamma D_\mu\Psi = -\bar{\Psi}\gamma^\mu D_\mu\Psi, \quad (D_\mu\bar{\Gamma}\bar{\Psi})\gamma^\mu\Gamma\Psi = -(D_\mu\bar{\Psi})\gamma^\mu\Psi,$$

hence $K[\Gamma\Psi] = -K[\Psi]$. Written out,

$$K = \frac{1}{2} e^\mu{}_a (\bar{\Psi}\gamma^a\partial_\mu\Psi - \partial_\mu\bar{\Psi}\gamma^a\Psi) + \frac{1}{4} e^\mu{}_a \omega_{\mu bc} \bar{\Psi}\{\gamma^a, S^{bc}\}\Psi,$$

a sum of coefficient functions of the gravitational field times bilinears with an odd number of gammas; each such bilinear changes sign by Lemma 2, whatever the coefficients are.

Step 2 (the Lagrangian). $U(S)$ is unchanged because S is, so

$$\mathcal{L}_{m,U}[\Gamma\Psi] = \sqrt{|g|} [-K - mS - U(S)] = -\sqrt{|g|} [K - (-m)S - (-U)(S)] = -\mathcal{L}_{-m,-U}[\Psi],$$

which for $U = \frac{\lambda}{2}S^2$ is (T1a).

Step 3 (the field equations). $\gamma^\mu D_\mu(\Gamma\Psi) = \gamma^\mu \Gamma D_\mu \Psi = -\Gamma \gamma^\mu D_\mu \Psi$, so

$$E_{m,\lambda}[\Gamma\Psi] = -\Gamma \gamma^\mu D_\mu \Psi - (m + \lambda S)\Gamma\Psi = -\Gamma[\gamma^\mu D_\mu \Psi - (-m - \lambda S)\Psi] = -\Gamma E_{-m,-\lambda}[\Psi],$$

and Γ is invertible, which gives (T1b). For the Euler-Lagrange expressions themselves, in any field and whatever their explicit form: (T1a) is an identity between functionals of Ψ , and $\Psi \mapsto \Gamma\Psi$ is a constant invertible linear substitution, so by the chain rule the Euler-Lagrange expressions of $\mathcal{L}_{m,\lambda}$ at $\Gamma\Psi$, multiplied by Γ , equal minus those of $\mathcal{L}_{-m,-\lambda}$ at Ψ ; they vanish together.

Step 4 (energy-momentum tensor and current). $T^\nu{}_\mu$ is the sum of $\delta^\nu_\mu \mathcal{L} / \sqrt{|g|}$ and kinetic bilinears with one gamma. By Step 2 with $m \rightarrow -m$, $\mathcal{L}_{-m,-\lambda}[\Gamma\Psi] = -\mathcal{L}_{m,\lambda}[\Psi]$, and by Step 1 each kinetic bilinear changes sign; hence $T^{(-m,-\lambda)}[\Gamma\Psi] = -T^{(m,\lambda)}[\Psi]$. The same holds for the tensor of the vielbein variation, whose additional term $\nabla_\lambda(\bar{\Psi}\{\gamma^\lambda, \Sigma^{\nu\rho}\}\Psi)$ contains three gammas. J^μ is a multiple of $\bar{\Psi}\gamma^\mu\Psi$ and changes sign by Lemma 2, and so does $Q = \int \sqrt{|g|} J^{x_4} d^7x$. This is (T1c).

Step 5 (the pair). Adding (T1c) to the quantities of Ψ gives (T1d).

Step 6 (statistics). $\Gamma = \text{diag}(-I_8, I_8)$ multiplies each component by $+1$ or -1 ; no Grassmann factors are reordered, so Steps 1 to 5 hold verbatim for anticommuting components, as identities between Grassmann-algebra elements.

The proof uses only Lemmas 1 and 2 and the linearity of $\mathcal{L}$, T and J in the coefficient functions $\sqrt{|g|}$, $e^\mu{}_a$ and $\omega_{\mu ab}$; hence T1 holds in every gravitational field (H1). QED.

4.4 T1 in the block form of the author's field

In the author's field the proof can be read off the block equations of section 2.5. $\Gamma(\psi_-, \psi_+) = (-\psi_-, \psi_+)$, and C is block diagonal (a product of four gammas; the algebra record gives $C = \text{diag}(-\sigma, \sigma)$ with σ the notebook's 8×8 involution), so $S = -\psi_-^\dagger \sigma \psi_- + \psi_+^\dagger \sigma \psi_+$ is unchanged and $V = m + \lambda S$ becomes $-m - \lambda S = -V$ for the parameters $(-m, -\lambda)$. The two block equations become

$$\sum_a f_a^{-1} \bar{\tau}^a \partial_a \psi_+ + 3H\psi_+ = (-V)(-\psi_-), \quad -\left(\sum_a f_a^{-1} \tau^a \partial_a \psi_- + 3H\psi_-\right) = -V\psi_+,$$

the same equations. None of the gravitational terms ($e^{\mp a_4} \sin^{-1/6} z$, $\tan z$, $3H$) changes.

Wolfram records (author's field)	sympy records (author's field)
T1_Lagrangian_primordial_commuting (408 monomials), T1_Lagrangian_primordial_grassmann (392 monomials), T1_field_equation_covariance_primordial_commuting, T1_field_equation_covariance_primordial_grassmann, T1_energy_momentum_primordial_commuting, T1_energy_momentum_primordial_grassmann, T1_current_primordial_commuting, T1_current_primordial_grassmann	C_equals_notebook_sigma16, T1.metric.commuting.lagrangian, T1.metric.grassmann.lagrangian, T1.metric.commuting.euler_lagrange_map, T1.metric.grassmann.euler_lagrange_map, T1.metric.commuting.pair_total_emt_zero, T1.metric.grassmann.pair_total_emt_zero

4.5 Remarks on T1

- Negative controls: $\mathcal{L}_{m,\lambda}[\Gamma\Psi] + \mathcal{L}_{m,\lambda}[\Psi] \neq 0$ and $\mathcal{L}_{m,\lambda}[\Gamma\Psi] + \mathcal{L}_{-m,\lambda}[\Psi] \neq 0$: the mass AND the coupling must change sign. For $\lambda \neq 0$, T1 is therefore not a pure $+m / -m$ pairing; for $\lambda = 0$ (free fields) it pairs $+m$ with $-m$ exactly.
- The partner's energy density, pressures and equations of state: $\rho' = -\rho$ and $p'_\mu = -p_\mu$ in every direction; for homogeneous on-shell states $\rho = mS + \frac{\lambda}{2}S^2$ and $p = \frac{\lambda}{2}S^2$ become $\rho' = -mS - \frac{\lambda}{2}S^2$ and $p' = -\frac{\lambda}{2}S^2$ (with $S' = S$). Every ratio of components, in particular $w_3 = p_3/\rho$, $w_t = p_t/\rho$ and $w_8 = p_8/\rho$, is the same for both members.
- T1 is not a symmetry of one theory: it changes the parameters and the sign of the action. Because $-\mathcal{L}$ and $\mathcal{L}$ have the same Euler-Lagrange equations, $\Gamma\Psi$ is a solution of the $(-m, -\lambda)$ theory; nothing more is asserted.
- Representation content: Γ is -1 on one irreducible $\text{Spin}(4,4)$ half and $+1$ on the other (section 2.2); the partner differs from Ψ by the relative sign of its two inequivalent $\text{Spin}(4,4)$ components, and Γ is itself an element of $\text{Pin}(4,4)$ (a product of eight unit vectors).

Wolfram records (negative controls)	sympy records (negative controls)
T1_Lagrangian_negative_controls_primordial_commuting, T1_Lagrangian_negative_controls_primordial_grassmann, T1_Lagrangian_negative_controls_diagonal8_commuting, T1_Lagrangian_negative_controls_diagonal8_grassmann, T1_Lagrangian_negative_controls_pointwise_commuting, T1_Lagrangian_negative_controls_pointwise_grassmann	T1.metric.commuting.negative_controls, T1.metric.grassmann.negative_controls

5 Theorem T2: the mirror pairing (both fields)

5.1 Hypotheses

- (H5) n is a space-like frame direction (x_1, x_2, x_3 or x_8), and $P_n = \Gamma\gamma^n$ is the $\text{Pin}(4,4)$ lift of the reflection R_n , of character -1 (Lemma 3).
- (H6) General-field version: the frame is reflected, $e' = R_n e$ (the same metric, since $R_n \eta R_n = \eta$, and the same $\sqrt{|g|}$), with its canonical connection, and $\Psi'(x) = \Gamma P_n \Psi(x) = \gamma^n \Psi(x)$ at the same point.
- (H7) Author's-field version: the map $\phi : x_8 \mapsto \pi/(6H) - x_8$ ($z \mapsto \pi - z$) takes the patch $z \in (0, \pi/2)$ onto the mirror patch $z \in (\pi/2, \pi)$; each patch carries its own positive vielbein ($e^8 = |\cot z| dx_8$, i.e. $-\cot z dx_8$ on the mirror patch); and $\Psi'(\phi(x)) = \gamma^8 \Psi(x)$. The brane $z = \pi/2$ is a degenerate surface of the metric ($g_{88} = \cot^2 z = 0$ and $\sqrt{|g|} = \cos z = 0$ there); extending the field to the mirror patch is the ASSUMED Z2 construction of Revision /SPEC.md section 7.
- (H8) Both statistics; $U(S) = \frac{\lambda}{2} S^2$ or any even function of S .

5.2 Statement

Theorem T2. Under (H5) to (H8), for both fields:

- (T2a) $\mathcal{L}_{m,\lambda}[\gamma^n \Psi; R_n e] = +\mathcal{L}_{-m,\lambda}[\Psi; e]$ in every gravitational field; in the author's field $\mathcal{L}_{m,\lambda}[\Psi'](\phi(x)) = \mathcal{L}_{-m,\lambda}[\Psi](x)$.
- (T2b) $S' = -S$, and $E_{m,\lambda}[\gamma^n \Psi; R_n e] = -\gamma^n E_{-m,\lambda}[\Psi; e]$: Ψ solves the $(-m, \lambda)$ equations if and only if its image solves the (m, λ) equations (in the author's field: on the patch and on the mirror patch respectively).
- (T2c) In the general-field version $T'_{\mu\nu} = T_{\mu\nu}^{(-m,\lambda)}[\Psi]$ and $J'^\mu = J^\mu[\Psi]$. In the author's field $T_{\mu\nu}[\Psi'](\phi(x)) = (R_8 T^{(-m,\lambda)}[\Psi](x) R_8)_{\mu\nu}$ and $J'^\mu = (R_8)^\mu{}_\nu J^\nu$: the energy density and every component without an x_8 index are EQUAL, not opposite. T2 pairs $+m$ with $-m$ at equal energy-momentum.
- (T2d) For a time-like n (character $+1$) the same construction gives $\mathcal{L} \rightarrow -\mathcal{L}_{-m,-\lambda}$, a T1-type map; the character -1 is what makes T2 a pairing $(m, \lambda) \rightarrow (-m, \lambda)$ with $\mathcal{L} \rightarrow +\mathcal{L}$.

5.3 Proof in a general field (frame reflection)

Step 1 (geometry). $g' = e'^T \eta e' = e^T R_n \eta R_n e = g$. The canonical connection of e' is $R_n \omega R_n$: R_n is a constant Lorentz transformation, and the canonical connection $\omega_\mu{}^a{}_b = e^a{}_\nu (\partial_\mu e^\nu{}_b + \Gamma^\nu{}_{\mu\lambda} e^\lambda{}_b)$ is built from the vielbein, its inverse and the Christoffel symbols $\Gamma^\nu{}_{\mu\lambda}$ of the unchanged metric (the symbol with indices is the Christoffel symbol, without indices the chirality matrix). By Lemma 3, $\Omega'_\mu = \frac{1}{2}(R_n \omega_\mu R_n)_{ab} S^{ab} = \gamma^n \Omega_\mu (\gamma^n)^{-1}$, so $D'_\mu(\gamma^n \Psi) = \gamma^n D_\mu \Psi$.

Step 2 (adjoint). By Lemma 3, $(\gamma^n)^T C = -\eta_{nn} C (\gamma^n)^{-1}$, hence $\bar{\Psi}' = \Psi^\dagger (\gamma^n)^T C = -\eta_{nn} \bar{\Psi} (\gamma^n)^{-1}$ and $D'_\mu \bar{\Psi}' = -\eta_{nn} (D_\mu \bar{\Psi}) (\gamma^n)^{-1}$.

Step 3 (gammas). The reflected frame gives $\gamma'^\mu = e^\mu_b (R_n)^b_a \gamma^a$, and Lemma 3 gives $\gamma'^\mu \gamma^n = -\gamma^n \gamma^\mu$. Therefore

$$\bar{\Psi}' \gamma'^\mu D'_\mu \Psi' = -\eta_{nn} \bar{\Psi} (\gamma^n)^{-1} \gamma'^\mu \gamma^n D_\mu \Psi = \eta_{nn} \bar{\Psi} \gamma^\mu D_\mu \Psi, \quad \bar{\Psi}' \Psi' = -\eta_{nn} \bar{\Psi} \Psi,$$

and in the same way $(D'_\mu \bar{\Psi}') \gamma'^\mu \Psi' = \eta_{nn} (D_\mu \bar{\Psi}) \gamma^\mu \Psi$, so $K' = \eta_{nn} K$ and $S' = -\eta_{nn} S$.

Step 4 (Lagrangian). For space-like n ($\eta_{nn} = +1$): $K' = K$ and $S' = -S$, so

$$\mathcal{L}_{m,\lambda}[\Psi'; e'] = \sqrt{|g|} [K + mS - \frac{\lambda}{2} S^2] = \mathcal{L}_{-m,\lambda}[\Psi; e],$$

which is (T2a). For time-like n : $K' = -K$ and $S' = S$, so $\mathcal{L}_{m,\lambda}[\Psi'; e'] = -\mathcal{L}_{-m,-\lambda}[\Psi; e]$, which is (T2d).

Step 5 (field equations). For space-like n ,

$$E_{m,\lambda}[\Psi'; e'] = \gamma'^\mu D'_\mu (\gamma^n \Psi) - (m + \lambda S') \gamma^n \Psi = -\gamma^n \gamma^\mu D_\mu \Psi - (m - \lambda S) \gamma^n \Psi = -\gamma^n E_{-m,\lambda}[\Psi; e],$$

which is (T2b).

Step 6 (energy-momentum tensor and current). The kinetic bilinears $\bar{\Psi} \gamma_\mu D_\nu \Psi$ with free indices transform like the contracted one in Step 3 (factor $\eta_{nn} = +1$), and $\mathcal{L}/\sqrt{|g|}$ transforms by Step 4, so $T'_{\mu\nu} = T_{\mu\nu}^{(-m,\lambda)}[\Psi]$; $\bar{\Psi}' \gamma'^\mu \Psi' = \eta_{nn} \bar{\Psi} \gamma^\mu \Psi$ gives $J' = J$. This is (T2c) in the general-field version.

Step 7 (statistics). γ^n is a constant signed permutation matrix; the substitution never reorders Grassmann factors, so Steps 1 to 6 hold for both statistics. QED.

5.4 Proof in the author's field (the Z2 mirror)

ϕ is an isometry of the author's metric: its components depend on x_8 only through $\sin z$ and $\cot^2 z$, both invariant under $z \mapsto \pi - z$. The positive vielbein of the mirror patch, pulled back by ϕ , is $R_8 e$: its hidden leg is $-\cot(\pi - z) d(\pi/(6H) - x_8) = -\cot z dx_8 = -e^8$, and the other legs are unchanged. $\sqrt{|g|} d^8 x$ is invariant ($|\cos z|$ is invariant and the Jacobian has modulus 1). Pulling the mirror-patch Lagrangian back by ϕ therefore gives the frame-reflected Lagrangian of section 5.3 with $n = x_8$, which proves (T2a) and (T2b) in the author's field; the coordinate components of T and J pick up the Jacobian R_8 of ϕ , which gives (T2c). QED.

In block form γ^8 has the blocks $\bar{\tau}^8 = \tau^8 = I_8$, so $\gamma^8(\psi_-, \psi_+) = (\psi_+, \psi_-)$: the mirror image exchanges the two inequivalent $\text{Spin}(4,4)$ halves, and $S = -\psi_-^\dagger \sigma \psi_- + \psi_+^\dagger \sigma \psi_+$ changes sign.

Wolfram records (author's field)	sympy records (author's field)
T2_mirror_is_isometry, T2_mirror_Lagrangian_commuting, T2_mirror_Lagrangian_grassmann, T2_mirror_energy_momentum_and_current_commuting, T2_mirror_energy_momentum_and_current_grassmann, T2_mirror_Euler_Lagrange_commuting, T2_mirror_Euler_Lagrange_grassmann	geometry.mirror_isometry, geometry.bbrane_degenerate, T2.metric.commuting.euler_lagrange_map, T2.metric.grassmann.euler_lagrange_map, T2.metric.commuting.emt, T2.metric.grassmann.emt, T2.metric.commuting.current, T2.metric.grassmann.current, T2.metric.commuting.S_odd, T2.metric.grassmann.S_odd

5.5 The eight reflections

The sympy checker also shows that neither the plain mirror ($P = 1$) nor Γ alone gives a relation $\mathcal{L}_{\text{mirror}} = \pm \mathcal{L}_{\text{patch}}$ with $(m, \lambda) \rightarrow (\pm m, \pm \lambda)$, that $P_8 = \Gamma \gamma^8$ alone gives $\mathcal{L} \rightarrow -\mathcal{L}_{m, -\lambda}$, and that its composition with Γ gives T2. The frame reflections of all eight directions, from the data of the pairing theory record (the Wolfram pointwise frame-reflection checks of every direction and the sympy frame-reflection checks of the general diagonal field, both statistics; the record names are in the T2 tables of section 9.2):

direction	η_{nn}	character of P_n	S under γ^n	kinetic term	map
x_1	+1	-1	$-S$	$+K$	$(-m, \lambda), +\mathcal{L}$ (T2)
x_2	+1	-1	$-S$	$+K$	$(-m, \lambda), +\mathcal{L}$ (T2)
x_3	+1	-1	$-S$	$+K$	$(-m, \lambda), +\mathcal{L}$ (T2)
x_4	-1	+1	$+S$	$-K$	$(-m, -\lambda),$ $-\mathcal{L}$ (T1-type)
x_5	-1	+1	$+S$	$-K$	$(-m, -\lambda),$ $-\mathcal{L}$ (T1-type)
x_6	-1	+1	$+S$	$-K$	$(-m, -\lambda),$ $-\mathcal{L}$ (T1-type)
x_7	-1	+1	$+S$	$-K$	$(-m, -\lambda),$ $-\mathcal{L}$ (T1-type)
x_8	+1	-1	$-S$	$+K$	$(-m, \lambda), +\mathcal{L}$ (T2)

In the author's field only the reflection of x_8 is realised by an isometry that maps the patch onto another patch (the mirror across the Z2 brane); the reflections of x_1, x_2, x_3 are frame statements.

6 The quantum-level reading (dirac16complex)

6.1 Hypotheses

- (HQ1) dirac16complex is canonically quantised with x_4 as the evolution time. The x_4 -derivative kernel of $\mathcal{L}_{m,\lambda}$ is $\frac{i}{2}\sqrt{|g|} N$ with $N = B$ (since $C\gamma^4 = iB$), and the canonical anticommutator on a slice $x_4 = \text{const}$ is $\{\Psi_A(x), \Psi_B^\dagger(y)\} = (N^{-1})_{AB} \delta^7(x-y)/\sqrt{|g|} = B_{AB} \delta^7(x-y)/\sqrt{|g|}$.
- (HQ2) dirac16complex00 is a classical field and has no quantum reading.

6.2 Statements

- (Q1) The chirality image $\chi = \Gamma\Psi$ carries the Krein metric $-B$: $\{\chi_A, \chi_B^\dagger\} = (\Gamma B \Gamma)_{AB} \delta/\sqrt{|g|} = -B_{AB} \delta/\sqrt{|g|}$, which is also the anticommutator demanded by its own Lagrangian $\mathcal{L}_{m,\lambda}[\Gamma\chi] = -\mathcal{L}_{-m,-\lambda}[\chi]$.
- (Q2) The image's own generators (energy, momentum and charge from its own Lagrangian) coincide with those of Ψ : the pair $\Psi, \Gamma\Psi$ is ONE quantum system, and the T1 identity $T^{(-m,-\lambda)}[\Gamma\Psi] = -T^{(m,\lambda)}[\Psi]$ is an identity between operators of that one system.
- (Q3) An independently quantised $(-m, -\lambda)$ universe has the anticommutator $+B$ (from $\mathcal{L}_{-m,-\lambda}$), its own vacuum and its own generators; it cannot be identified with $\Gamma\Psi$ ($B \neq -B$); on the product state space the generators add, $P_{\text{total}} = P_1 \otimes 1 + 1 \otimes P_2$, and no cancellation $P_1 + P_2 = 0$ follows.
- (Q4) The one-particle ($\lambda = 0$) Hamiltonians satisfy $\Gamma h_m \Gamma = h_{-m}$ (flat space, and a general field at a point with frozen coefficients) and $\gamma^8 h_m(k) \gamma^8 = h_{-m}(R_8 k)$: the $+m$ and $-m$ one-particle spectra are IDENTICAL, not opposite. In flat 4+4 space $h_m(k) = -im\gamma^4 - \gamma^4 \sum_{a \neq 4} k_a \gamma^a$ and $h_m(k)^2 = w^2 I_{16}$ with $w^2 = m^2 + k_1^2 + k_2^2 + k_3^2 + k_8^2 - k_5^2 - k_6^2 - k_7^2$; for every REAL frequency ($w^2 > 0$, with or without extra-time momentum) each of the two eigenspaces has dimension 8 and Krein inertia (4,4), the same for $+m$ and $-m$, and for imaginary or zero frequency (the growing extra-time modes) the eigenspaces are Krein-neutral (B vanishes on them).
- (Q5) The T2 image $\gamma^8 \Psi$ keeps the anticommutator $+B$: the mirror $(-m, \lambda)$ universe is an ordinary, independently quantisable copy with equal energies.

6.3 Proofs

(Q1) By Lemma 2, $\{\Gamma\Psi, (\Gamma\Psi)^\dagger\} = \Gamma\{\Psi, \Psi^\dagger\}\Gamma^\dagger = \Gamma B \Gamma \delta/\sqrt{|g|} = -B \delta/\sqrt{|g|}$. For the image's own Lagrangian, the x_4 -velocity term $\frac{i}{2}\sqrt{|g|} \chi^\dagger \Gamma B \Gamma \partial_4 \chi$ has the kernel $N = -B$, hence the anticommutator $N^{-1} = -B$: the two agree.

(Q2) At $\chi = \Gamma\Psi$ the image Lagrangian $\mathcal{L}_{m,\lambda}[\Gamma\chi]$ is $\mathcal{L}_{m,\lambda}[\Psi]$, the same function of the same operators, so its Legendre energy density and its Noether densities coincide with those of Ψ ; they are minus the generators of the $(-m, -\lambda)$ Lagrangian evaluated at χ . With the metric $-B$ and the generator density $-h_m$ the image evolves by $(-B)(-h_m) = B h_m$, exactly the (m, λ) dynamics.

(Q3) The Lagrangian $\mathcal{L}_{-m,-\lambda}[\chi_2]$ of an independent universe has the kernel $N = +B$, hence $\{\chi_2, \chi_2^\dagger\} = +B \delta/\sqrt{|g|}$, while $\{\Gamma\Psi_1, (\Gamma\Psi_1)^\dagger\} = -B \delta/\sqrt{|g|}$; and independence requires $\{\Psi_1, \chi_2^\dagger\} = 0$,

whereas $\{\Gamma\Psi_1, \Psi_1^\dagger\} = \Gamma B/\sqrt{|g|}$ has rank 16. So no identification $\chi_2 = \Gamma\Psi_1$ exists. On the product space the generators of the two systems add; the T1 identity relates operators of universe 1 only and gives no relation $P_2 = -P_1$. At zero momentum the block one-particle generator $\text{diag}(Bh_m, Bh_{-m})$ has the eigenvalues $-m$ and m , each 16 times, all nonzero for $m \neq 0$.

(Q4) $\Gamma\gamma^4\Gamma = -\gamma^4$ and $\Gamma\gamma^4\gamma^a\Gamma = \gamma^4\gamma^a$ give $\Gamma h_m\Gamma = h_{-m}$. With $(\gamma^8)^2 = 1$,

$$\gamma^8\gamma^4\gamma^8 = -\gamma^4, \quad \gamma^8\gamma^4\gamma^a\gamma^8 = \gamma^4\gamma^a \ (a \neq 4, 8), \quad \gamma^8\gamma^4\gamma^8\gamma^8 = -\gamma^4\gamma^8,$$

which gives $\gamma^8 h_m(k) \gamma^8 = h_{-m}(R_8 k)$. Similar matrices have equal spectra. The square and the eigenspaces are computed exactly; the samples of `Revision/pairing/pairing-theory.json` are:

m	momentum ($k_1, \dots, k_8$)	w	dimensions of the $+w$ and $-w$ eigenspaces	Krein inertia on $+w$ and on $-w$
2	(1, 2, 0, 0, 0, 0, 0, 4)	5	8 and 8	(4,4) and (4,4)
-2	(1, 2, 0, 0, 0, 0, 0, 4)	5	8 and 8	(4,4) and (4,4)
3	(0, 0, 0, 0, 0, 0, 0, 0)	3	8 and 8	(4,4) and (4,4)
-3	(0, 0, 0, 0, 0, 0, 0, 0)	3	8 and 8	(4,4) and (4,4)
1	(1, 0, 0, 0, 0, 0, 0, 0)	$\sqrt{2}$	8 and 8	(4,4) and (4,4)
-1	(1, 0, 0, 0, 0, 0, 0, 0)	$\sqrt{2}$	8 and 8	(4,4) and (4,4)
2	(0, 0, 0, 0, 1, 0, 0, 0)	$\sqrt{3}$	8 and 8	(4,4) and (4,4)
-2	(0, 0, 0, 0, 1, 0, 0, 0)	$\sqrt{3}$	8 and 8	(4,4) and (4,4)

The entry k_4 is not used (h_m contains no $\gamma^4\gamma^4$ term). The last two rows have an extra-time momentum and a real frequency. The general statement is proved, not sampled: (i) for real $w > 0$ the projectors $P_\pm = (1 \pm h_m/w)/2$ satisfy $P_+^\dagger B P_- = 0$ and $P_+^\dagger B P_+ = B P_+$ (from $Bh_m = h_m^\dagger B$ and $h_m^2 = w^2$), so the two eigenspaces are B -orthogonal and B is nondegenerate on each; (ii) in the good sector $[B, h_m] = 0$ and $\text{tr } B = \text{tr}(Bh_m) = 0$, so $\text{tr}(B P_\pm) = 0$ and B has inertia (4,4) on each eigenspace; (iii) the region $w^2 > 0$ is connected and contains the good sector (lowering k_5, k_6, k_7 to zero only increases w^2), and $P_\pm$ depend continuously on k there, so the inertia (4,4) holds at every real frequency. For imaginary w , $hu = wu$ and $hv = wv$ give $(w - \bar{w})u^\dagger Bv = 0$: the eigenspace is Krein-neutral; for $w = 0$, $h^2 = 0$ with rank 8 and $\ker h = \text{ran } h$ is B -neutral. Exact samples: $m = 1, k_5 = 2$ ($w = \pm i\sqrt{3}$); $m = 1, k_1 = 1, k_5 = 2$ ($w = \pm i\sqrt{2}$); $m = 1, k_5 = 1$ ($w = 0$): every eigenspace has dimension 8 and the form $u^\dagger Bv$ vanishes on it identically. In a general field at a point the same similarity $\Gamma h_m \Gamma = h_{-m}$ holds for symbolic momenta; in the pointwise test field $(\gamma^{x_4})^2 = g^{x_4 x_4} I_{16} = -\frac{2269482}{1990921} I_{16}$.

(Q5) Lemma 4 gives $\gamma^8 B (\gamma^8)^\dagger = +B$. QED.

statement	Wolfram records	sympy records
(HQ1)	first_order_form _and_anticommutator	canonical_antico mmutator_B, Q.canoni cal_anticommutator
(Q1)	Q_Krein_metric_o f_images, Q_symplic ctic_kernel_commuting, Q_symplectic_ker nel_grassmann	Q.image_krein_metric, Q.image_own_quan tisation
(Q2)	Q_generators_of_ the_image_commuting, Q_generators_of_ the_image_grassmann	Q.image_generato rs_same_dynamics
(Q3)	Q_no_identificat ion_of_independe nt_universes	Q.no_cancellatio n_independent_universes
(Q4)	Q_one_particle_f lat_dispersion, Q_one_pa rticle_maps, Q_one_pa rticle_Krein_signatures, Q_one_particle_K rein_inertia_rea l_frequencies, Q_one_pa rticle_complex_a nd_zero_frequenc ies_Krein_neutral, Q_one_particle_g eneral_field	Q.one_particle_maps, Q.one_particle_K rein_inertia, Q.one_pa rticle_Krein_ine rtia_proof, Q.one_pa rticle_complex_f requency_Krein_neutral, compare.theory.o ne_particle
(Q5)	Q_Krein_metric_o f_images	Q.T2_image_keeps_B

Relation to the good sector of the theory record. With $\Psi^\dagger$ realised as the Hilbert adjoint, the canonical anticommutator forces an indefinite (Krein) inner product (record `no_positive_inner_product`). The theory record constructs, in the good sector without extra-time momentum, a positive Fock representation in which $\Psi^\dagger$ is realised as χB with χ the Hilbert adjoint (records `good_sector_posi` `tive_fock_realisation` and `Fock_space_good_sector_example`). The statements Q1 to Q5 are statements about the canonical anticommutator; they neither use nor establish a positive-norm Fock space for either universe.

7 Corollary C1: a T1 pair as the source of the field equations for a_4

Hypotheses. (i) The gravitational field is the author's metric with the Einstein-Lovelock equations $\sum_{k=1}^3 \alpha_k E_{(k)}{}^\mu{}_\nu + \Lambda \delta^\mu_\nu = \kappa T^\mu{}_\nu$ of `Revision/SPEC.md` section 5. (ii) The complete source is the sum of the classical energy-momentum tensors of the two members of a T1 pair at the same points of the same patch (`dirac16complex00`, or `dirac16complex` as a Grassmann-algebra-valued classical bilinear). (iii) Nothing else sources the metric.

Statement. The total source vanishes identically, so a_4 must solve the vacuum equations. (a) In

Einstein gravity ($\alpha_1 = 1, \alpha_2 = \alpha_3 = 0$) there is no real solution for $H > 0$ and any Λ : the x_4 and x_8 components give $36H^2 + 2\Lambda = 0$ and $6(a'_4)^2 + 6H^2 = 0$. (b) In Einstein-Lovelock gravity the linear member $a_4 = AHx_4 + a_0$ needs $V = 0$, with

$$V = \alpha_1 - 40\alpha_2H^2 - 8A^2\alpha_2H^2 + 360\alpha_3H^4 + 144A^2\alpha_3H^4 + 72A^4\alpha_3H^4;$$

in Einstein-Gauss-Bonnet gravity ($\alpha_1 = 1, \alpha_3 = 0$) this gives $A^2 = (1 - 40\alpha_2H^2)/(8\alpha_2H^2)$, real if and only if $0 < \alpha_2H^2 \leq 1/40$, with $\Lambda = -\sum_k \alpha_k E_{(k)}^{x_4 x_4}$.

Proof. (T1d) gives $T + T' = 0$, so the right-hand side of the field equations is zero, and (a) and (b) are the vacuum statements of `Revision/field_equations_a4` (records below). QED.

statement	Wolfram records	sympy records
(a)	einstein_no_vacuum_solution	einstein_no_vacuum
(b)	linear_member_vacuum_factor, einstein_gauss_bonnet_vacuum_linear	linear_member_vacuum_factor, einstein_gauss_bonnet_vacuum_linear

Consequence (exact, under (i) to (iii)): a T1 pair cannot by itself be the source of the author's metric in Einstein gravity. This is a statement about sources in one common geometry; it is not a derivation that such a geometry, or such a pair, is created. For a T2 pair the sources do not cancel: the mirror partner carries the pulled-back energy-momentum tensor $R_8 T R_8$ of the $(-m, \lambda)$ configuration. For two independently quantised universes C1 does not apply (Q3).

8 Theorem T3: the Kohn-Sham level (dirac16complex)

`Revision/SPEC.md` section 9 asks for T3: at the Kohn-Sham level, the block map with the transformed boundary conditions maps the instantaneous Kohn-Sham problem with (m, λ) to the one with $(-m, +\lambda)$ with equal energies. It is proved by two independent exact verifiers in `Revision/pairing/kohn_sham/` (theorem record `Revision/pairing/kohn_sham/t3-theory.json`).

8.1 Hypotheses

- (H3.1) The problem: the instantaneous Kohn-Sham problem of `Revision/kohn_sham/ks-theory.json` at a fixed slice $a_{4,0} = a_4(x_4)$ of the author's metric: good sector (no extra-time momentum), a lattice of 3-space momenta k , and in each of the eight 2×2 blocks the Hamiltonian $h_j = j[-i\sigma_1 d/dy + M_{\text{eff}}(y)\sigma_2 + \kappa(y)k\sigma_3] + v_v(y)$ on $y \in [-L, 0]$ ($j = \pm 1$; four blocks (s_2, s_3) per j).
- (H3.2) The functional: Hartree plus the exact uniform-gas exchange, $M_{\text{eff}} = m + \frac{15}{16}\lambda S$, $v_v = -\lambda n/16$, $e_{\text{int}} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2$, no correlation; the energy-momentum tensor of that record.
- (H3.3) The brane (ASSUMED): the Z2 mirror (orbifold) at $y = 0$; both brane parities, $\chi_2(0) = 0$ and $\chi_1(0) = 0$, are solved and filled together.

- (H3.4) The tip (chosen): $(1 - Q(\theta))\chi(-L) = 0$ with $Q(\theta) = \cos \theta \sigma_3 + \sin \theta \sigma_2$.
- (H3.5) Mermin occupations at fixed particle number N and temperature T , with the filling convention of the Kohn-Sham record (positive branch of the $\lambda = 0$ problem plus the $k = 0$ zero modes, followed continuously in λ).
- (H3.6) The same H , L , $a_{4,0}$, lattice, extra-time volume, N and T for both members; quasi-free (mean-field) states.

8.2 Statement

For every self-consistent Kohn-Sham state with (m, λ, θ) , the map $(\chi, j) \rightarrow (\sigma_2 \chi, -j)$ at the same k , which is the chirality Γ of the 16-component orbital in the block basis (up to the phase s_2 per block), with the two brane parities exchanged, gives a self-consistent Kohn-Sham state with $(-m, +\lambda, \pi - \theta)$, and conversely (the map is an involution). The two states have the same levels with their degeneracies, the same occupations, chemical potential, particle number and entropy, EQUAL Kohn-Sham energy E_{KS} , grand potential and free energy, and equal energy-momentum profiles $\rho(y)$, $p_3(y)$, $p_t(y)$, $p_8(y)$; the densities transform as $n \rightarrow n$, $S \rightarrow -S$, $Q \rightarrow -Q$, so $M_{\text{eff}} \rightarrow -M_{\text{eff}}$ and $v_v \rightarrow v_v$. With the untransformed tip angle the spectra differ. Inside the ASSUMED Z2 orbifold the mirror copy of a self-consistent state carries $(-m, +\lambda)$.

8.3 Proof

1. $\sigma_2 h_j(M, k, v) \sigma_2 = h_{-j}(-M, k, v)$ as differential operators, and for the shooting form $\chi' = N\chi$, $N = M\sigma_3 - \kappa k \sigma_2 + ij(\varepsilon - v)\sigma_1$: $\sigma_2 N_j(M) \sigma_2 = N_{-j}(-M)$. So $\sigma_2 \chi$ is an eigen-orbital of block $-j$ with $-M$ at the same level ε .
2. $\sigma_2 Q(\theta) \sigma_2 = Q(\pi - \theta)$ and $\sigma_2(1 - \sigma_3) \sigma_2 = 1 + \sigma_3$: the tip condition goes into the one with $\pi - \theta$ (the canonical $\theta = 0$ into $\theta = \pi$) and the brane parities are exchanged; the boundary current $-ij \phi^\dagger \sigma_1 \chi$ and the norm are invariant, so the image problem is self-adjoint with the same normalisation.
3. Per orbital, n_o and t_o are invariant and s_o , q_o change sign; hence $n(y)$ is unchanged and $S(y)$ changes sign. $M_{\text{eff}}[-m, +\lambda, -S] = -M_{\text{eff}}[m, \lambda, S]$, v_v and e_{int} are even in S : the self-consistency loop commutes with the map exactly for $(-m, +\lambda)$ (not for $(-m, -\lambda)$).
4. The map is a bijection between the eigen-orbitals of the two problems with equal levels (both j and both parities belong to each problem), so occupations, μ , N and entropy agree; every term of E_{KS} , Ω , F and of the energy-momentum profiles is a sum of invariant densities, or contains the product $M_{\text{eff}} s_o$ of two odd factors, plus the even e_{int} . QED.

Independent confirmations inside the same records: for $k = 0$, $v = 0$ and constant M the characteristic functions of $(M, j, \theta = 0)$ and $(-M, -j, \theta = \pi)$ agree parity by parity, so the two spectra are equal level by level, while the untransformed tip gives a different characteristic function; the 2×2 map is the 16-component Γ in the block basis V of the Kohn-Sham record; the Z2 mirror $P_A \chi(y) = \sigma_3 \chi(-y)$ makes the doubled problem symmetric exactly when the mirror copy carries $(-m, +\lambda)$.

step	Wolfram records	sympy records
1	T3_block_hamilto nian_map, T3_ode_map	T3.block_hamilto nian_map, T3.ode_map
2	T3_tip_condition_map, T3_brane_paritie s_exchanged	T3.tip_condition_map, T3.brane_paritie s_exchanged
3	T3_orbital_densities, T3_mean_field_map	T3.orbital_densities, T3.mean_field_map
4	T3_energies_and_ emt_profiles_equal	T3.energies_and_ emt_profiles_equal
confirmations	T3_exact_k0_spectra, T3_Gamma_is_the_ block_map, T3_z2_mi rror_copy_carrie s_minus_m_plus_lambda	T3.exact_k0_spectra, T3.Gamma_is_the_ block_map, T3.z2_mi rror_copy_carrie s_minus_m_plus_lambda
comparison with the theorem record, numerical confirmation		compare.t3_theor y.theorem, compare. t3_theory.not_es tablished, T3.rust_ selftest_numeric al_confirmation

The ingredients recorded earlier by the Kohn-Sham theory reports agree with these steps: `block_Gamma_map`, `blocks_relation_to_Gamma`, `bc_mirror_map_PA`, `bc_mirror_parities_of_densities` and `bc_brane_parity_conditions` (both Kohn-Sham theory reports).

8.4 Numerical confirmation and interpretation

A numerical self-test of the Rust solver, labelled in its report as NOT a proof of T3: $(m, \lambda, \text{tip } \theta = 0)$ and $(-m, \lambda, \text{tip } \theta = \pi)$, solved independently, give the same sorted levels, occupations, Kohn-Sham energies and energy-momentum integrals and opposite S (worst deviation 9.95e-14, tolerance 1e-9). For $N = 8$, $\lambda = 0.01946$ and $a_{4,0} = 1$: $E_{KS} = -9.868426190876\text{e-}4$ versus $-9.868426190868\text{e-}4$ (80 levels), while the negative control with the untransformed tip $b(-L) = 0$ gives $E_{KS} = -2.0145243719\text{e}0$. For $N = 136$ and $\lambda = 0.0009298$: $E_{KS} = 3.239294915318\text{e}1$ in both runs (160 levels), negative control $2.9283751318\text{e}1$. The units are those of `Revision/kohn_sham/ks-theory.json`.

Interpretation (labelled, not a theorem): in its parameters the Kohn-Sham-level map is of the T2 type, $(m, \lambda) \rightarrow (-m, +\lambda)$ at equal energies, not of the T1 type, although its 16-component matrix is the chirality Γ of T1: the Kohn-Sham expectation values are taken with the fixed Krein metric B of the canonical quantisation, under which Γ flips the sign ($\Gamma B \Gamma = -B$, statement Q1), so $S \rightarrow -S$ at the Kohn-Sham level while $S[\Gamma\Psi] = S[\Psi]$ for the classical bilinear.

What T3 does not establish: no creation process, rate or amplitude; only instantaneous (adiabatic) mean-field states (the time-dependent problem is OPEN); the Z2 brane is ASSUMED and the tip angle must be transformed; no correlation; no statement about two independently quantised universes; no back-reaction (the Kohn-Sham energy-momentum tensor does not satisfy the source conditions of the a_4 equations, record `ks_profiles_violate_algebraic_condition`). The a_4 equations with the Kohn-Sham source (`Revision/field_equations_a4/ks_source/reports/ks`

-source-a4.json, 23 of 23 checks) confirm this exactly: the recorded states conserve energy-momentum along x_8 (B2_x8_conservation_on_every_profile), so their violation of $p_3 + p_t = 2p_8$ is exactly their dependence on x_8 , and it persists after averaging over the hidden direction (C1_averaged_algebraic_condition_fails); the coupled problem has no solution, so nothing about a_4 is derived from it, and the integration of a truncated, averaged system there is an approximation with residuals of order one, not used here.

8.5 Completion of T3 (2026-10-08)

An adversarial verification of T3 (2026-10-08) re-ran both T3 verifiers, which reproduce the theorem record `t3-theory.json` and the two T3 reports byte for byte, and re-derived every step of the proof; no error was found in the statement or the proof. Three gaps were found. A completion closes them by exact computation in a separate record, `t3-completion.json`, written by the Wolfram verifier `verify_t3_completion.wls` and re-derived independently by the sympy checker `check_t3_completion.py` (all in the folder `Revision/pairing/kohn_sham`; the reports lie in its subfolder `reports/`); T3 and its statements S1 to S5 are unchanged.

1. At the adversarial verification the Kohn-Sham coefficients $15/16$, $-1/16$, $15/32$, $-1/32$ of (H3.2) were written into the T3 verifier instead of being read from the Kohn-Sham record, and one of its clauses about v_v compared an expression with itself. The completion reads the potentials from `Revision/kohn_sham/ks-theory.json`, checks $M_{\text{eff}} - m = \partial e_{\text{int}}/\partial S$ and $v_v = \partial e_{\text{int}}/\partial n$, and repeats the map of step 3 with them ($M_{\text{eff}}[-m, +\lambda, -S] = -M_{\text{eff}}[m, \lambda, S]$; with $(-m, -\lambda)$ it fails, the control).
2. The filling convention (H3.5) was used for the $-m$ member without showing that its particle set is the image of the $+m$ member's. The completion shows that the map sends the $\lambda = 0$ levels with $\varepsilon > 0$ onto those of the image problem, commutes with the continuation in λ , and takes the $k = 0$ brane zero mode to the brane zero mode of the image in its own boundary conditions (odd parity, tip $\theta = \pi$); with the untransformed tip the image mode violates the tip condition (control). So the occupied set of the image is the image of the occupied set.
3. The proof covers the four diagonal energy-momentum components of the 2×2 reduction. Statement S6 extends it to the 16-component level: with the expectation rule $\rho = \sum f u u^\dagger B$ of the canonical quantisation and $\Gamma B \Gamma = -B$, the image state has $\rho' = -\Gamma \rho \Gamma$, so n is unchanged, S and Q change sign, and every component of the 16-component energy-momentum tensor and of the current J^a of the $(-m, +\lambda)$ image state equals that of the original state.

gap	Wolfram records	sympy records
1	T3C_mean_field_coefficients_from_ks_theory	T3C.mean_field_coefficients_from_ks_theory
2	T3C_filling_convention_mapped	T3C.filling_convention_mapped
3 (S6)	T3C_krein_rule_16_component	T3C.krein_rule_16_component

gap	Wolfram records	sympy records
T3 reports, the two demonstrations, comparison with the completion record		T3C.t3_reports_pass, T3C.rust_demo_numerical_demonstration, T3C.reference_demo_numerical_demonstration, compare.t3_completion

The numerical demonstrations (not part of the proof) solve, at the slices $a_{4,0} = 0, 0.5, 1, 1.5, 2$ of the prescribed background history $a_4 = AHx_4$ ($A = 1$, the extra times deflating as e^{-a_4}), three members of each state: the $+m$ universe ($m, \lambda, \theta = 0$), the $-m$ universe ($-m, +\lambda, \theta = \pi$) and a control.

- Rust solver (report `t3-rust-demo.json`, 210 states: all 75 ground and 135 Mermin states of the canonical matrix of the Kohn-Sham record): the $+m$ members reproduce the committed canonical matrix to $2.538\text{e-}10$; the image equals the $+m$ state to $2.179\text{e-}13$ (ground) and $3.877\text{e-}12$ (thermal), tolerance $1\text{e-}9$, label by label with the sector map $(n_2, j, \text{parity}) \rightarrow (n_2, -j, \text{other parity})$, including the energy-momentum integrals and profiles, with S , Q , M_{eff} of opposite sign (`t3_equal_ground_states`, `t3_equal_thermal_states`). The shooting method is covariant under the map, so this equality is at the rounding level by construction; it tests the solver's handling of the transformed boundary conditions, the labels, the filling convention, the self-consistency and the Mermin root for both members. Controls: with the untransformed tip the density profile differs from the $+m$ state's by at least $4.750\text{e+}00$ of its maximum (`negative_control_untransformed_tip`; 196 states with a converged control; in 14 states, all with $\lambda < 0$, its self-consistent field does not converge, which is reported as such and is not a proof that no such state exists); the T1 parameters ($-m, -\lambda, \theta = \pi$) differ from the $+m$ state by at least $1.210\text{e-}02$ in all 150 states with $\lambda \neq 0$ (`negative_control_lambda_sign`): the Kohn-Sham partner carries $+\lambda$.
- Independent reference solver (report `t3-reference-demo.json`, 18 states, finite differences with Richardson extrapolation): the image equals the $+m$ state to $2.043\text{e-}14$ in the ground states after extrapolation (tolerance $1\text{e-}9$; the Mermin states pass the same test), while on a single grid the two discretisations differ by up to $2.23\text{e-}04$, so this equality is a continuum statement; with the untransformed tip the density profile differs by at least $5.296\text{e+}00$ of its maximum; the $-m$ universes of the two solvers agree to $1.255\text{e-}11$ (`reference_image_equals_rust_image`).

What the completion does not establish: the same limits as T3 (section 8.4); in addition the filling convention remains a CONVENTION whose justification is open, and S6 is an equality of expectation values in Kohn-Sham (quasi-free) states with the fixed Krein metric B , not a statement about two independently quantised universes (the list of what is not established in the completion record).

9 Verification records

9.1 Reports and their counts

Every check has a name, a verdict and a detail. Counts at the time of writing (the publication test re-reads them):

report	checks	PASS	FAIL
Revision/pairing /reports/wolfram- pairing.json	101	101	0
Revision/pairing /reports/python- pairing.json	66	66	0
Revision/pairing /kohn_sham/repor ts/wolfram-t3.js on	10	10	0
Revision/pairing /kohn_sham/repor ts/python-t3.json	13	13	0
Revision/pairing /kohn_sham/repor ts/wolfram-t3-co mpletion.json	3	3	0
Revision/pairing /kohn_sham/repor ts/t3-rust-demo. json	7	7	0
Revision/pairing /kohn_sham/repor ts/t3-reference- demo.json	7	7	0
Revision/pairing /kohn_sham/repor ts/python-t3-com pletion.json	7	7	0
Revision/algebra /reports/wolfram -algebra.json	45	45	0
Revision/algebra /reports/python- algebra.json	35	35	0
Revision/theory/ reports/wolfram- field-theory.json	84	84	0
Revision/theory/ reports/python-f ield-theory.json	70	70	0

report	checks	PASS	FAIL
Revision/field_equations_a4/reports/wolfram-a4-report.json	52	52	0
Revision/field_equations_a4/reports/python-a4-report.json	63	63	0
Revision/field_equations_a4/ks_source/reports/ks-source-a4.json	23	23	0
Revision/kohn_sham/reports/ks-theory-wolfram.json	46	46	0
Revision/kohn_sham/reports/ks-theory-python.json	58	58	0
Revision/kohn_sham/reports/ks-rust-solver.json	42	42	0

The pairing theorems are verified in four gravitational fields: the author’s metric on the patch $z \in (0, \pi/2)$ (primordial); the same metric on the mirror patch; a general diagonal (4,4) field $e^a_\mu = h_a(x_1, \dots, x_8)\delta^a_\mu$ with eight arbitrary functions of all eight coordinates (diagonal8); and a general non-diagonal field at one point (pointwise), with exact rational e^a_μ and $\partial_\rho e^a_\mu$, $\det e = 1411/6561$, a non-diagonal metric and 448 of the 512 connection components nonzero; plus the linearity (kernel) argument that covers every field. The canonical connection of each test field satisfies the vielbein postulate exactly. The Wolfram side and the sympy side share no code; the sympy side re-builds the gammas from the author’s tau formulas and compares them entry by entry with `Revision/algebra/gammas.json`.

9.2 Checks per theorem

The tables below list all 101 checks of the Wolfram pairing report and all 66 checks of the sympy pairing report (section 9.1), grouped by theorem; every one has the verdict PASS (the checks of T3 are listed in section 8.3). Among the common sympy checks, the three comparison checks for T1, T2 and Q record that the sympy side confirms each Wolfram theorem independently (with 17, 13 and 10 named sympy checks), and the comparison check for the limits records that the two independently written lists of what the theorems do not establish cover the same topics; section 11.2 merges them.

T1, Wolfram (51 checks):

check	check	check
T1_kernel_scalar	T1_kernel_kinetic	T1_kernel_connection

check	check	check
T1_kernel_field_equation	T1_linearity_argument	T1_scalar_and_kinetic_primordial_commuting
T1_Lagrangian_primordial_commuting	T1_Lagrangian_negative_controls_primordial_commuting	T1_field_equation_covariance_primordial_commuting
T1_Euler_Lagrange_derived_primordial_commuting	T1_Euler_Lagrange_map_primordial_commuting	T1_energy_momentum_primordial_commuting
T1_current_primordial_commuting	T1_general_potential_primordial_commuting	T1_scalar_and_kinetic_primordial_grassmann
T1_Lagrangian_primordial_grassmann	T1_Lagrangian_negative_controls_primordial_grassmann	T1_field_equation_covariance_primordial_grassmann
T1_Euler_Lagrange_derived_primordial_grassmann	T1_Euler_Lagrange_map_primordial_grassmann	T1_energy_momentum_primordial_grassmann
T1_current_primordial_grassmann	T1_scalar_and_kinetic_diagonal8_commuting	T1_Lagrangian_diagonal8_commuting
T1_Lagrangian_negative_controls_diagonal8_commuting	T1_field_equation_covariance_diagonal8_commuting	T1_Euler_Lagrange_derived_diagonal8_commuting
T1_Euler_Lagrange_map_diagonal8_commuting	T1_energy_momentum_diagonal8_commuting	T1_current_diagonal8_commuting
T1_scalar_and_kinetic_diagonal8_grassmann	T1_Lagrangian_diagonal8_grassmann	T1_Lagrangian_negative_controls_diagonal8_grassmann
T1_field_equation_covariance_diagonal8_grassmann	T1_Euler_Lagrange_derived_diagonal8_grassmann	T1_Euler_Lagrange_map_diagonal8_grassmann
T1_energy_momentum_diagonal8_grassmann	T1_current_diagonal8_grassmann	T1_scalar_and_kinetic_pointwise_commuting
T1_Lagrangian_pointwise_commuting	T1_Lagrangian_negative_controls_pointwise_commuting	T1_field_equation_covariance_pointwise_commuting
T1_energy_momentum_pointwise_commuting	T1_current_pointwise_commuting	T1_scalar_and_kinetic_pointwise_grassmann
T1_Lagrangian_pointwise_grassmann	T1_Lagrangian_negative_controls_pointwise_grassmann	T1_field_equation_covariance_pointwise_grassmann
T1_energy_momentum_pointwise_grassmann	T1_current_pointwise_grassmann	Gamma_properties

T1, sympy (23 checks):

check	check	check
T1.general_field	T1.general_field	T1.general_field
.matrix_identities	.random_instance	.random_instance
	.commuting	.grassmann
T1.metric.commut	T1.metric.commut	T1.metric.commut
ing.S_invariant	ing.lagrangian	ing.negative_controls
T1.metric.commut	T1.metric.commut	T1.metric.commuting.emt
ing.euler_lagran	ing.euler_lagrange_map	
ge_derived		
T1.metric.commut	T1.metric.commut	T1.metric.grassm
ing.pair_total_emt_zero	ing.current	ann.S_invariant
T1.metric.grassm	T1.metric.grassm	T1.metric.grassm
ann.lagrangian	ann.negative_controls	ann.euler_lagran
		ge_derived
T1.metric.grassm	T1.metric.grassmann.emt	T1.metric.grassm
ann.euler_lagrange_map		ann.pair_total_emt_zero
T1.metric.grassm	T1.metric.commut	T1.metric.grassm
ann.current	ing.general_potential	ann.general_potential
T1.diagonal8.commuting	T1.diagonal8.grassmann	

T2, Wolfram (28 checks):

check	check	check
T2_Pn_in_Pin44	T2_Pn_covers_the	T2_character_of_Pn
	_reflection	
T2_Gamma_times_P	T2_kernels_gamma	T2_frame_reflect
n_is_gamma_n	_n_with_frame_re	ion_pointwise_x1
	flection	_commuting
T2_frame_reflect	T2_frame_reflect	T2_frame_reflect
ion_pointwise_x1	ion_pointwise_x2	ion_pointwise_x2
_grassmann	_commuting	_grassmann
T2_frame_reflect	T2_frame_reflect	T2_frame_reflect
ion_pointwise_x3	ion_pointwise_x3	ion_pointwise_x4
_commuting	_grassmann	_commuting
T2_frame_reflect	T2_frame_reflect	T2_frame_reflect
ion_pointwise_x4	ion_pointwise_x5	ion_pointwise_x5
_grassmann	_commuting	_grassmann
T2_frame_reflect	T2_frame_reflect	T2_frame_reflect
ion_pointwise_x6	ion_pointwise_x6	ion_pointwise_x7
_commuting	_grassmann	_commuting
T2_frame_reflect	T2_frame_reflect	T2_frame_reflect
ion_pointwise_x7	ion_pointwise_x8	ion_pointwise_x8
_grassmann	_commuting	_grassmann
T2_mirror_is_isometry	T2_mirror_Lagran	T2_mirror_energy
	gian_commuting	_momentum_and_cu
		rrent_commuting
T2_mirror_Euler_	T2_mirror_Lagran	T2_mirror_energy
Lagrange_commuting	gian_grassmann	_momentum_and_current_gr
		assmann
T2_mirror_Euler_		
Lagrange_grassmann		

T2, sympy (16 checks):

check	check	check
geometry.brane_d egenerate	geometry.mirror_ isometry	T2.general_field .matrix_identities
T2.general_field .reflection_table	T2.metric.commut ing.reflection_table	T2.metric.commut ing.euler_lagrange_map
T2.metric.commuting.emt	T2.metric.commut ing.current	T2.metric.commut ing.S_odd
T2.metric.grassm ann.reflection_table	T2.metric.grassm ann.euler_lagrange_map	T2.metric.grassmann.emt
T2.metric.grassm ann.current	T2.metric.grassm ann.S_odd	T2.diagonal8.fra me_reflections.c ommuting
T2.diagonal8.fra me_reflections.g rassmann		

Q, Wolfram (12 checks):

check	check	check
Q_Krein_metric_o f_images	Q_symplectic_ker nel_commuting	Q_generators_of_ the_image_commuting
Q_symplectic_ker nel_grassmann	Q_generators_of_ the_image_grassmann	Q_one_particle_f lat_dispersion
Q_one_particle_maps	Q_one_particle_K rein_signatures	Q_one_particle_K rein_inertia_rea l_frequencies
Q_one_particle_c omplex_and_zero_ frequencies_Krei n_neutral	Q_one_particle_g eneral_field	Q_no_identificat ion_of_independe nt_universes

Q, sympy (10 checks):

check	check	check
Q.canonical_anti commutator	Q.image_krein_metric	Q.image_own_quan tisation
Q.image_generato rs_same_dynamics	Q.no_cancellatio n_independent_universes	Q.one_particle_maps
Q.one_particle_K rein_inertia	Q.one_particle_K rein_inertia_proof	Q.one_particle_c omplex_frequency _Krein_neutral
Q.T2_image_keeps_B		

Common checks (gammas, geometry, comparison with the Wolfram record), sympy (17 checks):

check	check	check
gammas.clifford	gammas.C	gammas.Gamma
gammas.Gamma_ant	gammas.B	gammas.equal_wol
icommutates		fram_fixture
geometry.omega_a	geometry.gamma_c	geometry.spin_co
ntisymmetric	ovariantly_constant	nnection_components
compare.theory.status	compare.theory.r	compare.theory.k
	eflection_table	rein_signs
compare.theory.o	compare.theory.t	compare.theory.t
ne_particle	heorem_T1	heorem_T2
compare.theory.t	compare.theory.n	
heorem_Q	ot_established	

Common checks (fixture, test fields, connections, determinism), Wolfram (10 checks):

check	check	check
fixture_Clifford	fixture_definitions	C_and_B_basic
_relation		
primordial_vielbein	pointwise_field_	connection_primordial
	is_general	
connection_mirror_patch	connection_diagonal8	connection_pointwise
theory_file_dete		
rministic		

9.3 Re-verification for this document

For the first edition (2026-10-01) both pairing verifiers were re-run on a copy of the working tree: 99 of 99 and 64 of 64 checks passed and the outputs were byte-identical to the recorded files. After the review of the same day the pairing verifiers were extended (the Krein inertia of every real-frequency eigenspace, the Krein-neutral imaginary-frequency eigenspaces) and re-run on the working tree: `verify_pairing.wls` 101 of 101 PASS in about 80 s, `check_pairing.py` 66 of 66 PASS in about 140 s; the T3 verifiers 10 of 10 (about 3 s) and 13 of 13 (about 1 s). Two runs of each give byte-identical outputs. The completion records of section 8.5 (2026-10-08) are cited with the counts of their committed reports (Wolfram 3 of 3, sympy 7 of 7, the two demonstrations 7 of 7 each); their run times on the development machine, shared with other jobs, are about 15 s and 2 s for the completion verifiers, 10 to 15 min for the Rust demonstration (630 solver runs, 8 at a time) and about 10 min for the reference demonstration. The reports cited in this document are the files of the working tree.

10 Reproduction

From the repository root (the sympy pairing checker reads the Wolfram theory record for its final comparison, so the Wolfram verifier runs first; the Kohn-Sham theory checker reads the record written by the Kohn-Sham source-conditions checker, so that one runs before it, and so does the analysis of the a_4 equations with the Kohn-Sham source, which re-derives that record; the T3 completion runs after the T3 verifiers: the Wolfram completion verifier first, then the Rust

demonstration (it needs the built Rust solver), then the reference demonstration (it compares its $-m$ universe with the Rust table), and the sympy completion checker last (it reads the completion record and both demonstration reports); `<target>` is the cargo target folder and `<scratch>` an empty working folder; the PDF command is one command, the backslash is the line continuation of a POSIX shell):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/theory/wolfram/verify_field_theory.wls
python Revision/theory/python/check_field_theory.py
wolframscript -file Revision/pairing/wolfram/verify_pairing.wls
python Revision/pairing/python/check_pairing.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/kohn_sham/theory/verify_ks_theory.wls
python Revision/field_equations_a4/python/check_ks_source_conditions.py
python Revision/field_equations_a4/ks_source/ks_source_a4.py
python Revision/kohn_sham/theory/check_ks_theory.py
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3.wls
python Revision/pairing/kohn_sham/python/check_t3.py
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3_completion.wls
cargo build --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
python Revision/pairing/kohn_sham/numerics/t3_rust_demo.py \
    --solver <target>/release/revision_ks_solver --work <scratch> --jobs 8
python Revision/pairing/kohn_sham/numerics/t3_reference_demo.py --jobs 8
python Revision/pairing/kohn_sham/python/check_t3_completion.py
python scripts/build_provenance_pdf.py Revision/docs/PAIR_CREATION_PROOFS.md \
    --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_pair_creation_proofs_publication.py -v
```

The PDF build runs the Markdown-to-LaTeX builder twice (with different hash seeds), runs pdflatex three times into each of two fresh directories, requires warning-free logs and byte-identical PDFs, and checks the PDF against its entry `pair-creation-proofs` in `Revision/pdf-specifications.json` (the Revision registry; the registry of the earlier stages is not touched).

11 What is proved and what is not

11.1 Proved

Exactly, for both fields (dirac16complex with Grassmann components, dirac16complex00 with commuting components) unless stated, under the hypotheses stated with each result:

1. T1 (section 4): $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$; the solutions of the (m,λ) theory and of the $(-m,-\lambda)$ theory correspond one to one; $T \rightarrow -T$ and $J \rightarrow -J$; the pair has zero total energy-momentum, current and charge as classical bilinears; in every gravitational field taken as a fixed background.
2. T2 (section 5): Γ combined with a $\text{Pin}(4,4)$ reflection of character -1 gives $\mathcal{L}_{m,\lambda}[\gamma^n\Psi; R_n e] = +\mathcal{L}_{-m,\lambda}[\Psi; e]$; in the author's field the mirror across the Z2 brane $z = \pi/2$ (ASSUMED

construction) pairs a $(-m, \lambda)$ solution on the patch with an (m, λ) solution on the mirror patch at EQUAL energy-momentum and charge.

3. Q (section 6, dirac16complex only): the chirality image carries the Krein metric $-B$ and is the same quantum system re-labelled; an independently quantised $(-m, -\lambda)$ universe carries $+B$ and cannot be identified with it; the generators of two independent universes add without cancelling; the $+m$ and $-m$ one-particle ($\lambda = 0$) spectra are identical (flat space, or frozen coefficients at a point); the T2 image keeps $+B$.
4. C1 (section 7): a T1 pair as the complete classical source of the author's metric is a zero source; in Einstein gravity the author's metric then has no solution for $H > 0$.
5. T3 (section 8, dirac16complex): every self-consistent instantaneous Kohn-Sham state with (m, λ, θ) is mapped by the block map of Γ , with the brane parities exchanged and $\theta \rightarrow \pi - \theta$, onto a self-consistent Kohn-Sham state with $(-m, +\lambda, \pi - \theta)$ with equal levels, occupations, Kohn-Sham energy and energy-momentum profiles, and $S \rightarrow -S$ (ASSUMED Z2 brane); with the completion (section 8.5), at the 16-component level with the expectation rule of the canonical quantisation, every component of the energy-momentum tensor and of the current is unchanged and S, Q change sign (S6), and the filling convention of the $+m$ member is carried onto the $-m$ member.

11.2 Not established

1. No creation process: the theorems map solutions to solutions and quantities to quantities; nothing in these equations produces a universe, a pair of universes or a change of the number of universes, and no transition between “no universe” and “two universes”, no initial state, no vacuum decay and no tunnelling process is derived.
2. No rate, probability or amplitude: no transition amplitude, probability, cross-section, rate or Bogoliubov coefficient for creating universes of masses $\{+m, -m\}$ is computed or implied; no wave function of the universe and no path integral is part of these theorems.
3. No dynamical necessity: no equation and no conservation law forces the partner to exist; a single universe with mass $+m$ is an equally valid solution without its partner. T1 and T2 are correspondences between solutions of two parameter sets, not a mechanism.
4. T1 is not a symmetry of a single theory: it changes the parameters and the sign of the action, and for $\lambda \neq 0$ it pairs (m, λ) with $(-m, -\lambda)$, not $+m$ with $-m$ at fixed coupling. The pure pairing $(m, \lambda) \rightarrow (-m, \lambda)$ is T2, at equal, not opposite, energy-momentum: T2 does not reverse the sign of T .
5. The vanishing total energy-momentum and charge of a T1 pair holds for classical bilinears and as an operator identity within ONE quantum system; it does not hold for two independently quantised universes, whose generators add without cancelling.
6. Test-field statement: the gravitational field is fixed and the same for both members. The back-reaction through the field equations for a_4 is not part of the theorems; corollary C1 is the only statement about it, and it concerns the sum of classical sources in one common geometry.

7. The Z2 brane: the mirror across $z = \pi/2$ uses the ASSUMED Z2 construction; the metric is degenerate there ($g_{88} = 0$ and $\sqrt{|g|} = 0$), and no junction condition, brane tension or matching of the field across the brane is derived.
8. Quantum positivity: in flat 4+4 space every real-frequency eigenspace of the one-particle Hamiltonian has Krein inertia (4,4) (proved, section 6.3), and the eigenspaces of imaginary or zero frequency are Krein-neutral; a positive-norm Fock space for either universe is not established by these theorems (the good-sector construction of the theory record is a separate result, not used here).
9. dirac16complex00 is a classical field: no quantum statement is made for it.
10. The Kohn-Sham level T3 holds for the instantaneous (adiabatic) mean-field Kohn-Sham states only, with the ASSUMED Z2 brane and the transformed tip condition; the time-dependent Kohn-Sham problem is open, no correlation is included, and T3 says nothing about back-reaction (section 8); the history $a_4 = AHx_4$ of the numerical demonstrations is a prescribed background, the filling of the Kohn-Sham states is a CONVENTION, and the demonstrations of section 8.5 are numerical, not proofs.

11.3 The answer to the request

The request “PROVE that Universes of masses $\{+mass, -mass\}$ are created in pairs” is answered exactly as follows, for dirac16complex and for dirac16complex00. Proved: to every solution with mass m the explicit maps Γ (T1: with $\lambda \rightarrow -\lambda$, opposite energy-momentum and charge, in every gravitational field) and γ^8 with the ASSUMED Z2 mirror construction (T2: same λ , equal energy-momentum, in the author’s primordial gravitational field) assign a solution with mass $-m$; for dirac16complex, at the Kohn-Sham level, the block map of Γ with the transformed boundary conditions assigns to every self-consistent instantaneous Kohn-Sham state with (m, λ) one with $(-m, +\lambda)$ at equal energies (T3, ASSUMED Z2 brane). That universes occur in pairs is not proved: these are maps between the solutions of two parameter sets, and a single universe with mass $+m$ is an equally valid solution without its partner. At the quantum level of dirac16complex the chirality partner is the same quantum system with the Krein metric $-B$, and two independently quantised universes of masses $+m$ and $-m$ have identical one-particle ($\lambda = 0$) spectra (flat space, or frozen coefficients at a point) and no cancelling generators. Not proved: that such universes are CREATED, in pairs or otherwise. No creation process, rate or amplitude follows from these equations.